JEEVAN SHAJI V V

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ABOUT

At ThinkSeed Systems, I work as a Software Engineer specializing in automotive systems and real-time operating systems (RTOS). I design, develop, and implement high-performance embedded software solutions to enhance overall system reliability.

I actively contribute to next-generation digital key solutions on the vehicle side, leveraging Bluetooth Low Energy (BLE), Ultra-Wideband (UWB), and Near Field Communication (NFC) technologies.

I hold a B.Tech in Computer Science and Engineering with a Minor in Electronics and Communication Engineering from the Government College of Engineering, Kannur. With strong proficiency in C, C++, and Java, and a solid foundation in data structures and algorithms, I bring a performance-driven approach to automotive and embedded software development.

CORE SKILLS

- Communication Protocols: SPI, UART, I2C, CAN ..etc.
- Wireless Technologies: UWB (Ultra-Wideband), BLE (Bluetooth Low Energy), NFC (Near-Field Communication).
- Security Implementation: mbedTLS for encryption, secure boot, and authentication in embedded systems.
- Programming Languages: C/C++, Java, Python.
- Embedded Systems Development: Firmware development, debugging, and optimization for low-latency, high-reliability systems.
- Experience in RTOS and linux OS.
- Android App Development: Java, Android SDK, REST APIs, BLE communication.
- Tools Platforms: Embedded IDEs, mbedTLS, Git, Android Studio, VS Code..etc

EDUCATION

GOVERNMENT COLLEGE OF ENGINEERING kannur — Kannur, Kerala
Btech in computerscience and engineering — CGPA: 9.22/10.0 — 2023

ST JOSEPHS HIGHER SECONDARY SCHOOL — Thalassery, Kerala
Computer Science — 98.416 — 2019

WORK EXPERIENCE

THINKSEED SYSTEMS PVT LTD — WHITEFIELD, BANGALORE
SOFTWARE ENGINEER, Embedded System — Oct, 2023 – Ongoing

- Designed and implemented next-generation digital key solutions (CCC-compliant and custom implementation) leveraging Bluetooth Low Energy (BLE) and Ultra-Wideband (UWB), NFC technologies for secure and precise vehicle access.
- Engineered BLE communication frameworks for embedded systems, ensuring reliable wireless connectivity and low-latency data exchange.
- Implemented UWB applications for precise ranging and location tracking in embedded systems.
- Integrated multiple wired and wireless communication protocols — CAN, SPI, UART, and I^2C — to optimize data transmission and overall system performance.
- Hands on experience in linux OS and RTOS.
- Developed cryptographic security modules using MbedTLS for embedded platforms, enhancing data protection and communication integrity.

PROJECT WORK

Automated Plastic Waste Detection Bot — 2023

- Developed an autonomous robot for plastic waste detection and collection using deep learning and embedded technologies.
- Implemented a computer vision model to detect plastic waste with high accuracy, enhancing the efficiency of waste collection.
- Designed and integrated a robotic arm for automated waste grabbing and disposal, improving the automation process.
- Optimized real-time navigation using embedded systems, enabling the bot to autonomously search for and collect plastic waste.

Medcare — 2022

- Developed a hospital management web application using the MERN stack to streamline patient record management and token booking.
- Implemented a secure and scalable backend with MongoDB and Express.js, ensuring efficient data storage and retrieval.
- Designed an intuitive user interface with React.js for seamless interaction and improved patient experience.
- Integrated real-time token booking and patient record tracking, reducing manual workload for hospital staff.